

PCAP Analysis

URL: <https://app.letsdefend.io/challenge/pcap-analysis>

Walkthrough

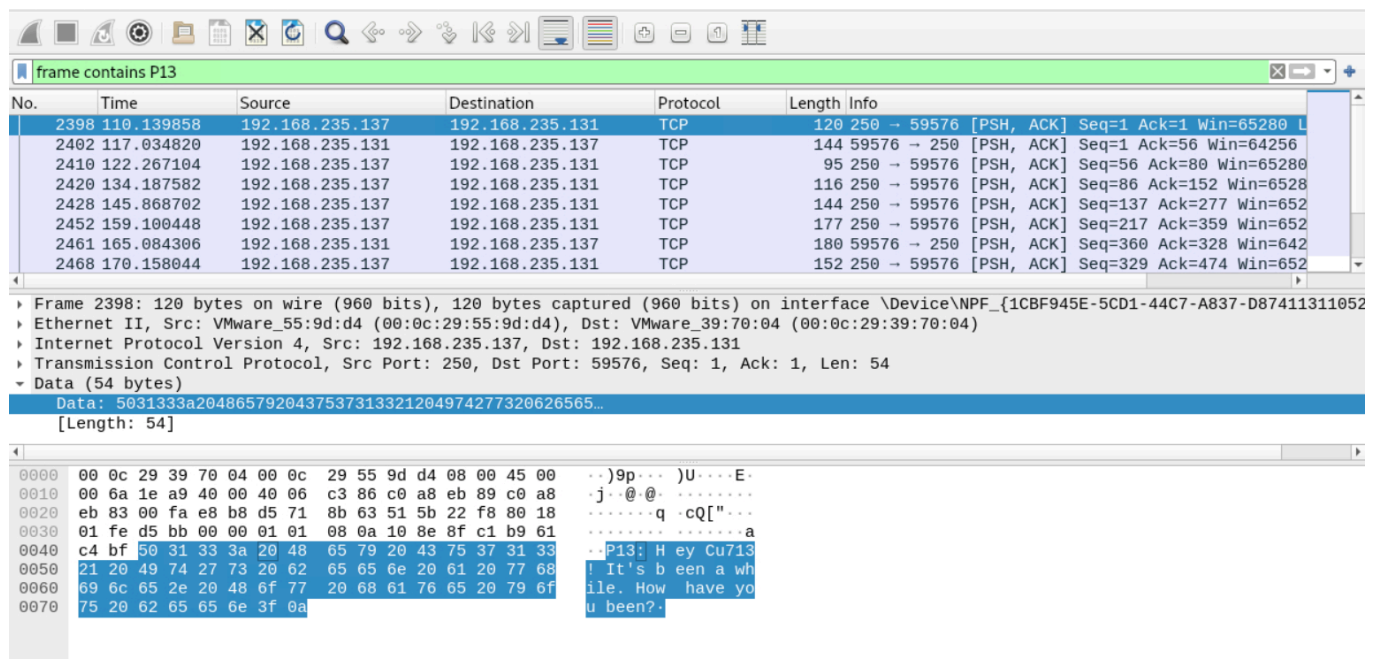
This walkthrough demonstrates how to analyze HTTP traffic in Wireshark to identify file uploads and extract key forensic details.

The investigation focuses on suspicious communication from a host labeled **P13**.

Question 1: In network communication, what are the IP addresses of the sender and receiver?

We are told that the investigation concerns **P13's computer traffic**, so let's begin with a filter:

frame contains P13



Wireshark packet capture analysis showing a list of packets filtered by "frame contains P13". The first packet (No. 2398) is selected, showing details for Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The data section shows a text message: "P13: Hey Cu713! It's been a while. How have you been?"

In the **Data** section of the first packet, we can see:

P13: Hey Cu713! It's been a while. How have you been?

It looks like we're on the right track.

Let's go to the top menu: **Statistics** → **Conversations** → **IPv4** (tab)

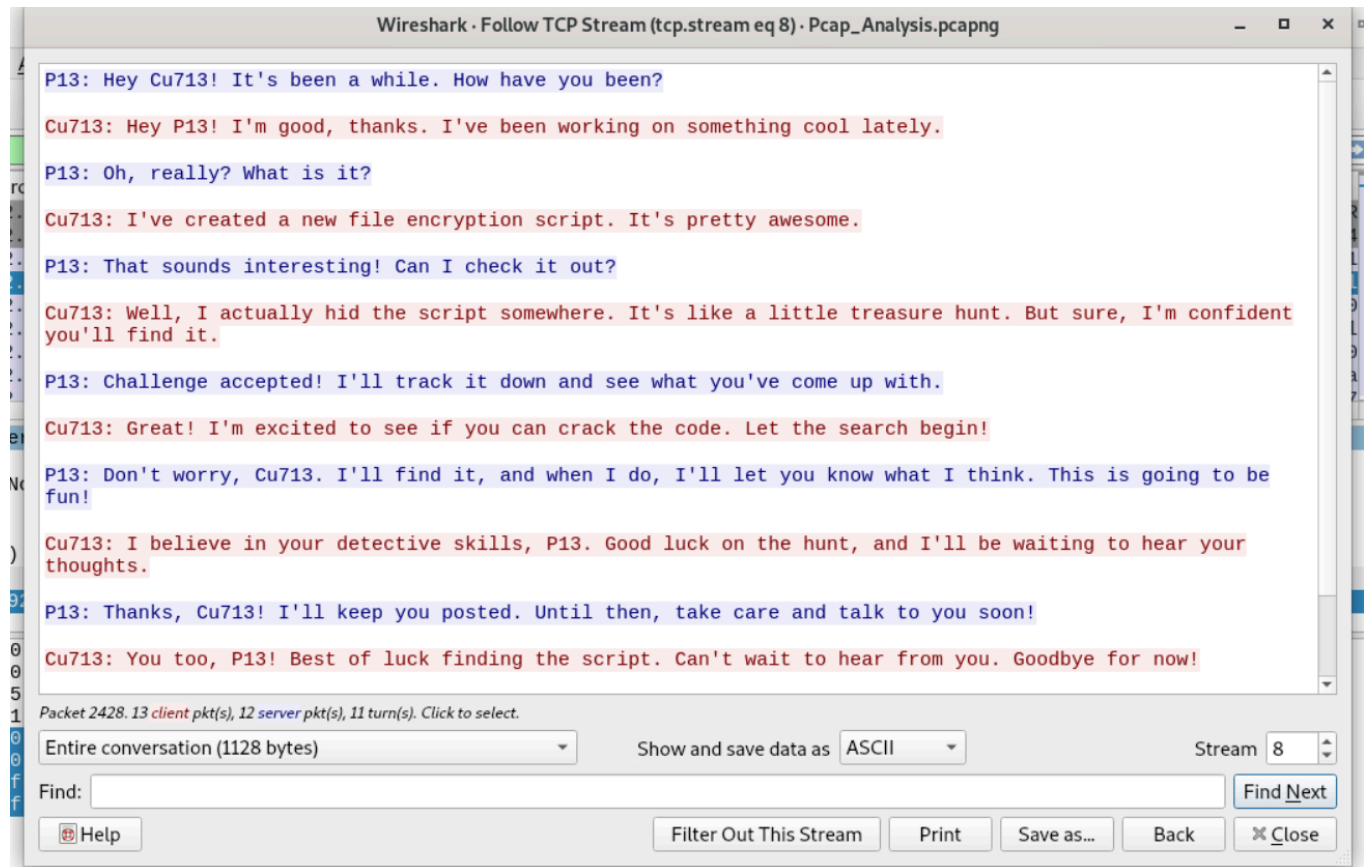
Make sure to select **Limit to display filter**.

Among the listed conversations, only one pair communicates in both directions:

- 192.168.235.131
- 192.168.235.137

Let's take a look at exactly how this conversation goes.

Go back to the main window and select the first packet sent between the above IP addresses. Then right-click on it, choose **Follow** and then **HTTP Stream**.



It is clear that this is a kind of chat and **P13** appears to be a nickname.

❓ In network communication, what are the IP addresses of the sender and receiver?

192.168.235.137, 192.168.235.131

Question 2: P13 uploaded a file to the web server. What is the IP address of the server?

We already know the IP address of P13 - 192.168.235.137 .

Now, we look for file upload activity.

Method 1: Filter POST requests:

```
ip.src == 192.168.235.137 && http.request.method == "POST"
```

You may notice multiple OCSP requests, which can be ignored in this case.

Note

OCSP (Online Certificate Status Protocol) is used to check the revocation status of digital certificates and is more efficient than CRLs.

Method 2: Filter multipart uploads

```
ip.src == 192.168.235.137 && http.content_type contains multipart
```

Now we can see a single packet - an HTTP POST request to `/panel.php`.

 **P13 uploaded a file to the web server. What is the IP address of the server?**

192.168.1.7

Question 3: What is the name of the file that was sent through the network?

Open the packet for `/panel.php` and follow the HTTP stream.

We can see the request headers:

```
POST /panel.php HTTP/1.1
Host: 192.168.1.7
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/112.0.5615.50 Safari/537.36
Accept-Encoding: gzip, deflate
Accept: */*
Connection: keep-alive
Content-Type: multipart/form-data; boundary=----
WebKitFormBoundary4BIlFVC7HcRhJCay
Cookie: PHPSESSID=a2kttlfov0k65t5g1c0s32lpit8
Content-Length: 107671

-----WebKitFormBoundary4BIlFVC7HcRhJCay
```

```
Content-Disposition: form-data; name="file"; filename="file"
Content-Type: =-----=
```

The relevant line with the filename is:

```
Content-Disposition: form-data; name="file"; filename="file"
```

🔗 What is the name of the file that was sent through the network?

file

Question 4: What is the name of the web server where the file was uploaded?

In order to find details about the web server, we need to see the server's response.

The easiest way is to scroll to the end of the HTTP stream or change the dropdown at the bottom to display only the server response (from **Entire conversation** to **192.168.1.7:80 -> 192.168.235.137:40740**).

```
HTTP/1.1 200 OK
Date: Tue, 23 May 2023 18:01:22 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.0.25
X-Powered-By: PHP/8.0.25
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Content-Length: 486
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=UTF-8

file uploaded at uploads/file
```

The header `Server` holds the answer to the question:

```
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.0.25
```

🔗 What is the name of the web server where the file was uploaded?

Apache

Question 5: What directory was the file uploaded to?

The web server's response also reveals the location of the uploaded file.

We can see in the response: `file uploaded at uploads/file`

🔍 What directory was the file uploaded to?

uploads

Question 6: How long did it take the sender to send the encrypted file?

Close the window with streams.

You should already have the stream filter applied.

```
tcp.stream eq 22
```

If not, just use this query.

Let's go again to the top menu: **Statistics** → **Conversations**.

Make sure to select **Limit to display filter** and check the value in the **Duration** column.

🔍 How long did it take the sender to send the encrypted file?

0.0073